

Network Adjustment

Workflow Documentation

The objective of network adjustment in a gravity survey is to transform relative gravity differences (observed with a relative-gravity meter) into absolute gravity values through known reference stations. This process determines the single, most probable best-fit gravity value at each station by integrating all available observations and their respective uncertainties (Kennedy et al, 2021).

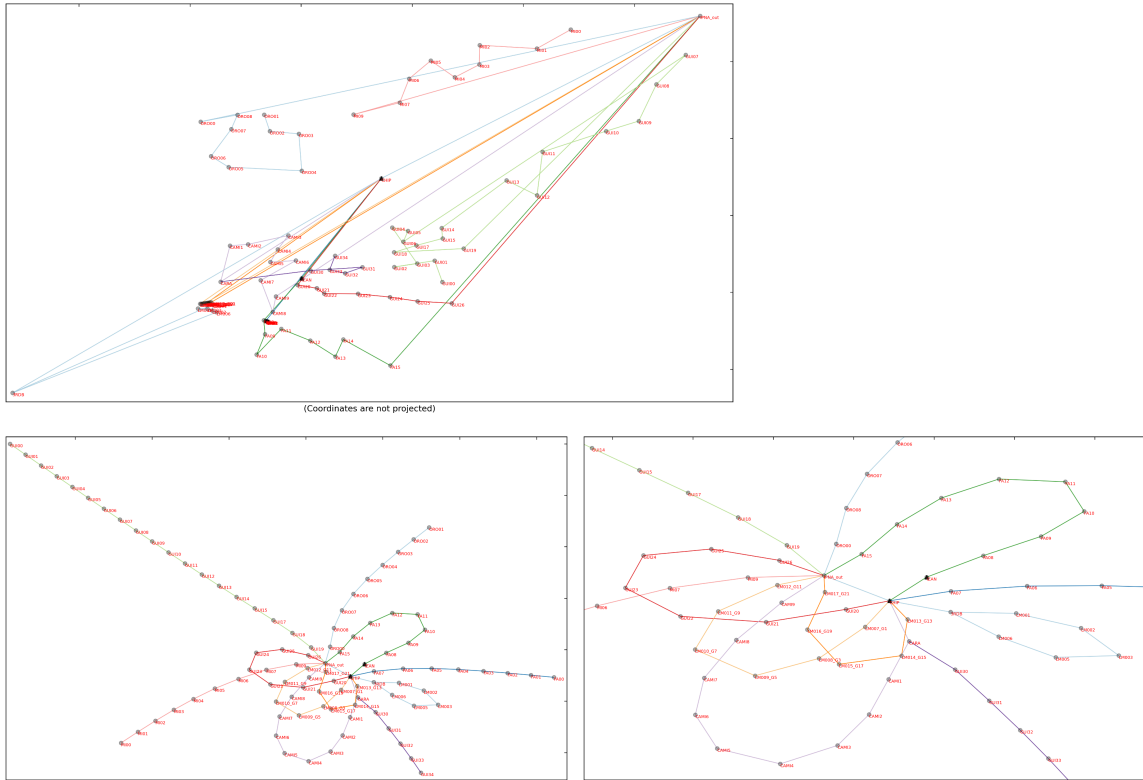
1. GSadjust Software Principles

The network adjustment was performed using GSadjust, a USGS software package designed for processing relative and absolute gravity data (Kennedy et al., 2021). The underlying principle is to take the gravity differences (ΔG) measured between stations and anchor them to the known values of the absolute reference stations. To achieve this, the software uses a Weighted Least Squares (WLS) algorithm, which finds the most probable gravity value for every station by minimizing the total error across the network. This method ensures that the most precise measurements (with the lowest standard deviations) have the greatest influence on the final adjusted result.

While GSadjust includes built-in functions for modeling instrumental drift during the adjustment process, these features were not utilized in this study as the drift had been previously resolved during the initial data reduction phase. This approach was specifically chosen due to the Star Network topology of the survey; since there were no station repetitions within individual loops to constrain an internal drift model, a dedicated external drift correction was performed prior to the simultaneous network adjustment to ensure maximum stability.

2. Network

The CSIC Survey consists of multiple open loops (11), forming a Star Network configuration centered on a primary pivot station, "IPNA_out" and a repeated reference station "CHIP". During each loop, this pivot station and/or the reference stations (CHIP, IZAN) are visited to ensure network connectivity, and thus being able to calculate absolute values from the reference datum constraints.



As we can see some loops are opened AND closed by pivot (b) or reference stations, while others are just a 'line' (a).

Month	Loop	Date	Ref stations / Pivots	Measured station	Tare applied (μGal)
April	0	2024-04-02 2024-04-03	3RDB (2) ; CHIP IPNA_out	LM001-006	
May	1	2024-05-03	CHIP	FA00-07	
	2	2024-05-06 2024-05-08	IPNA_out	GUI00-06 GUI07-19	-500
	3	2024-05-09	CHIP; IZAN; IPNA_out	FA08-15	-500
	4	2024-05-10	IPNA_out	MI00-09	-500
June	5	2024-06-03	CHIP; IPNA_out	GUI20-26	-915
July1	6	2024-07-08	CHIP; IPNA_out	LM007_G1 - 012_G11	
	7	2024-07-09	CHIP; IPNA_out	LM013_G13 - 017_G21	-250
July2	8	2024-07-31	CHIP; IPNA_out; CARA	CAMI1-9	
October	9	2024-10-07	CARA	GUI30-34	-422
	10	2024-10-13			+5982

		2024-10-14	IPNA_out	ORO00-08	
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Loop :

- yellow : open loop, only one ref station
- Green : circular loop : repeated ref stations

3. Reference Stations

The survey is anchored to three absolute gravity reference stations: CHIP, IZAN, and 3RDB. These stations belong to the Instituto Geográfico Nacional (IGN) network and provide the absolute gravity framework necessary to constrain the relative measurements.

Within the GSadjust environment, these stations function as fixed datums. By comparing the relative gravity differences (ΔG) to these known absolute values, the Weighted Least Squares algorithm can effectively distribute errors and solve for the instrument's calibration factor.

Station	Gravity (μGal)	Std. Dev. (μGal)	Date	Meas. Height (cm)	Gradient ($\mu\text{Gal}/\text{cm}$)
CHIP	978988902.2	12.4	2014-05-24	72	-3.086
3RDB	978962965.2	22.9	2014-05-25	72	-3.349
IZAN	978891479.8	2.3	2014-10-15	72	-4.365

In the end 3RDB was not used as a reference station. Instead it was treated as an unconstrained station to serve as an independent check of the network's accuracy.

The units are the correct to enter in GSadjust network Tab.

[Consulta los valores de la gravedad absoluta - Instituto Geográfico Nacional](#) - Select the base station and click on Reseña to get the PDF with all the informations. Also in [/03_Network_adjustment/Reference stations/](#)

4. Process

a. Pre-Process - Full relative gravity value

Compute the full relative gravity value because GSadjust just reads ObsG and doesn't do Dial + ObsG. The used scale factor here was 1 because we didn't find the calibration table of the gravimeter, and GSadjust calculates one that fits the data perfectly.

$$\text{Full Relative Gravity} = (\text{Dial} * \text{Scale Factor}) + \text{ObsG}$$

b. Data Tab

- **Load** raw Burris data, with already the exact expected column names.

[/GSadjust/FINAL_Data_for_GSadjust_BURRIS.csv](#)

- **Tide correction**: we kept the Tide correction (Longman model) already applied from the gravimeter, although GSadjust offers the Agnew model.
- **Average** : the software makes a gravity value average for each station (because we have multiple measurements at a few sec/min interval for better accuracy). It weights each measurement given the standard deviation and beam error, giving one relative value for each station with its standard deviation
- **Network** : divide the survey into different loops (see Network part). In the tree view, keep only 1 survey with different loops.

Name	Date	g (μGal)
3		
☒ CHIP (3)	2024-05-09	2522306.8 ± 0.6
☒ IZAN (1)	2024-05-09	2423422.9 ± 0.7
☒ FA08 (1)	2024-05-09	2483463.4 ± 0.9
☒ FA09 (1)	2024-05-09	2485799.0 ± 1.2
☒ FA10 (1)	2024-05-09	2507779.5 ± 0.3
☒ FA11 (1)	2024-05-09	2493772.8 ± 0.4
☒ FA12 (1)	2024-05-09	2569768.2 ± 1.4
☒ FA13 (1)	2024-05-09	2651808.5 ± 0.4
☒ FA14 (1)	2024-05-09	2646788.4 ± 1.2
☒ FA15 (1)	2024-05-09	2745807.0 ± 0.7
☒ IPNA_out (3)	2024-05-09	2834701.6 ± 0.3
4		
☒ MI00 (1)	2024-05-10	2796559.4 ± 0.4
☒ MI01 (1)	2024-05-10	2757560.8 ± 2.4
☒ MI02 (1)	2024-05-10	2785613.3 ± 0.8
☒ MI03 (1)	2024-05-10	2756597.1 ± 2.4
☒ MI04 (1)	2024-05-10	2749613.9 ± 1.3
☒ MI05 (1)	2024-05-10	2815618.8 ± 2.4
☒ MI06 (1)	2024-05-10	2819759.9 ± 1.4
☒ MI07 (1)	2024-05-10	2798773.5 ± 0.6
☒ MI09 (1)	2024-05-10	2813922.9 ± 1.0
☒ IPNA_out (4)	2024-05-10	2834748.7 ± 0.5
5		
☒ CHIP (4)	2024-06-03	2522109.2 ± 0.4
☒ GUI20 (1)	2024-06-03	2435156.0 ± 0.9
☒ GUI21 (1)	2024-06-03	2449124.3 ± 1.5
☒ GUI22 (1)	2024-06-03	2478042.8 ± 5.4
☒ GUI23 (1)	2024-06-03	2525115.4 ± 5.0
☒ GUI24 (1)	2024-06-03	2575059.4 ± 1.7
☒ GUI25 (1)	2024-06-03	2626030.1 ± 1.2
☒ GUI26 (1)	2024-06-03	2695965.1 ± 0.8
☒ IPNA_out (5)	2024-06-03	2834917.4 ± 0.7
6		

Figure : Tree view example is GSadjust, we see they finish by IPNA_out and measures a reference station.

- **Data check** : if the standard deviation of the value at one station is too high or red (<5 μGal), we can uncheck certain outlier measurements. Done with GUI00, GUI05, MI06, GUI33.

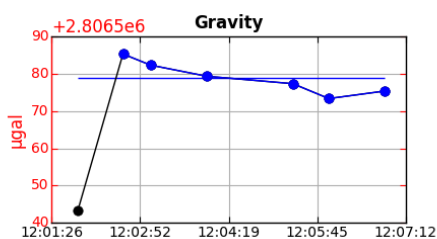


Figure : Example of GUI00 with its first gravity value outlier.

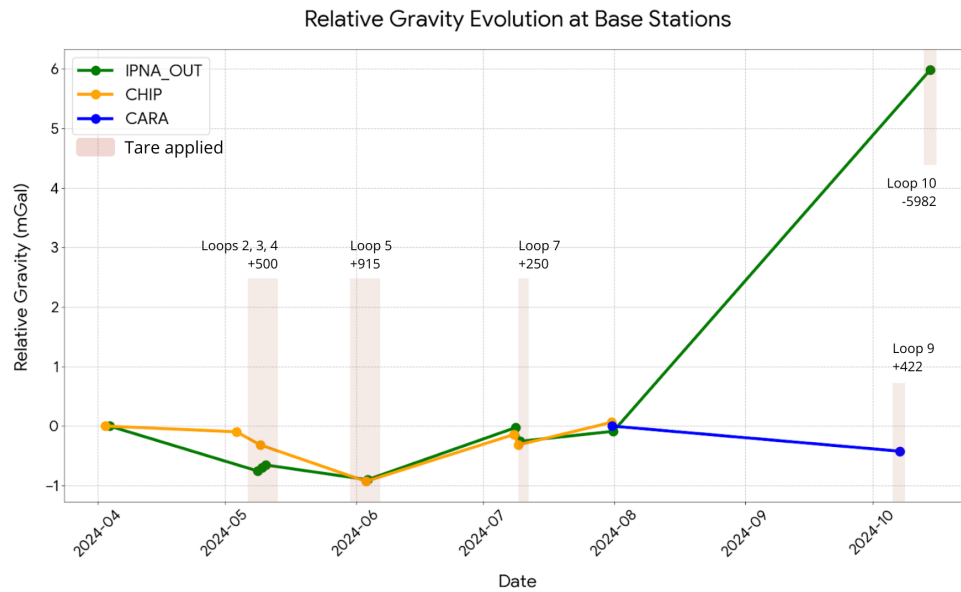
- **Delete 3RDB** from the 10/10/2024 measurement and from the datum values. It increases the final standard deviation and incoherence; Weakens the last loop; Creates more errors; Is on a single day so better not to take into account (could have other tares or drift).

c. Drift Tab

- **Drift** correction method : keep it on '**None**'. We Because there isn't any station repeat in the loops (star network) and we have already done short term (per monthly survey) drift correction with python.

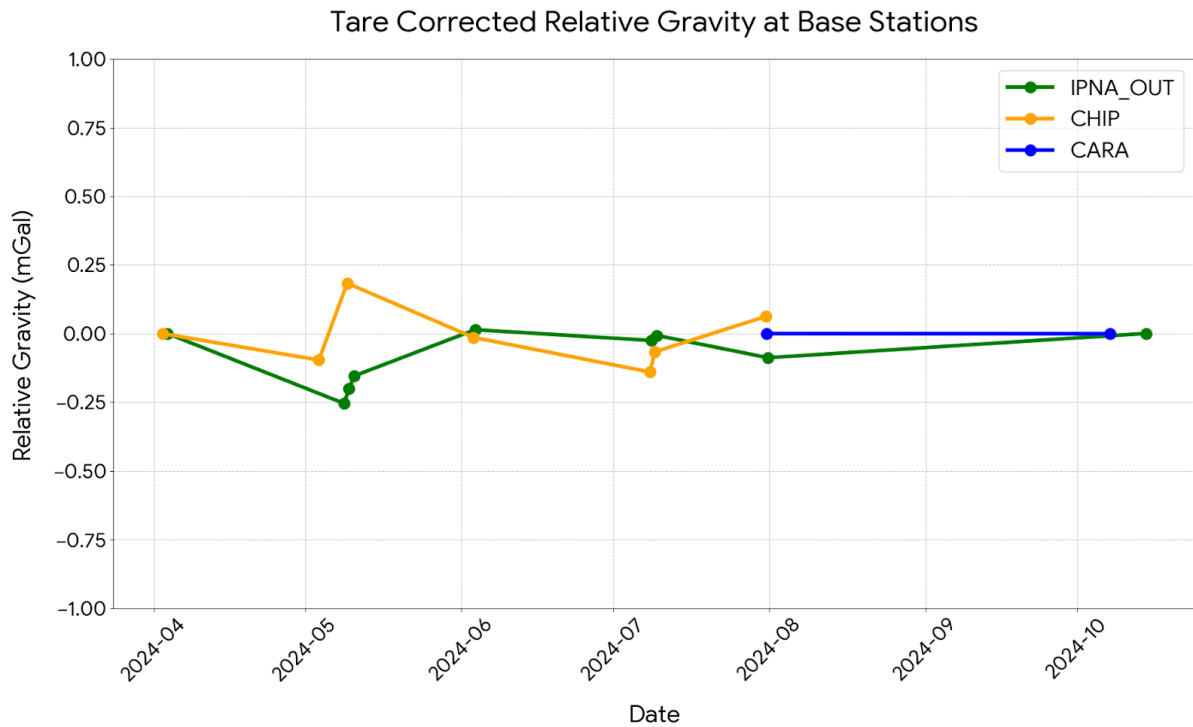
- **Tares & Offsets**

The instrument also undergoes a long term drift (between monthly surveys) and can sometimes show offsets which are jumps in the repeated station value supposed to be constant. Consequently we have to apply tares to realign these stations on a base to keep consistency and coherence in the network (see tare value applied in network table).



Select loop : Edit/add tare : select date (usually a few minutes before the first measurement needing tare to be sure it counts) / and tare value* in μGal .

*The value needs to be at the sign : there is 300 μGal missing, so you put -300 it will add 300 to the gravity value.



After the Tare Correction, the value of the repeated stations has only a 0.2 mGal variation (instead of 6 mGal before).

d. Network Adjustment

- Add the Datum : absolute references

Adjustment/Import abs.g (simple) / *Reference stations/datum_simple.txt*

Check that the units are correct, and enter manually the gradient and the measurement height.

- Import the DeltsG : *Adjustment / populate delta table*
- **Adjustment options :**

After a few try to obtain statistical validation and consistency, the options chosen for the network adjustment are :

- Minimum Standard Deviation (for adjustment) : 75 (μGal) → accounts for the instrument error
- Calculate relative meter calibration coefficient : 0.984657. This value is consistent with factory specifications for Burris meters (around 1), confirming instrument health.

Network adjustment options

☐ Standard deviation multiplier: pre-minimum 1.0000

☐ Standard deviation multiplier: post-minimum 1.0000

☐ Add to standard deviation: pre-minimum 0.0

☒ Minimum standard deviation 75.0

☒ Calculate relative meter calibration coefficient

☐ Specify calibration coefficient

	Meter	Calibration factor
1	B-28	0.984657

Significance level for global model test 0.05

Restore defaults Apply to current survey (2024-04-02) Apply to all surveys Cancel

The workspaces can be open : */Network Adjustment/OG3_no_3RDB.gsa* for the original form without calculations and */Network Adjustment/FINAL_network_adjustment.gsa*

5. Results and validation

As results, a few files can be used, starting by :

The summary textfile */Results_files/Final_adj_summary.txt*

The .csv absolute Gravity Values */Results_files/Final_adjustment.csv*

Shapefiles : .cpg ; .dbf ; .prj ; .shp

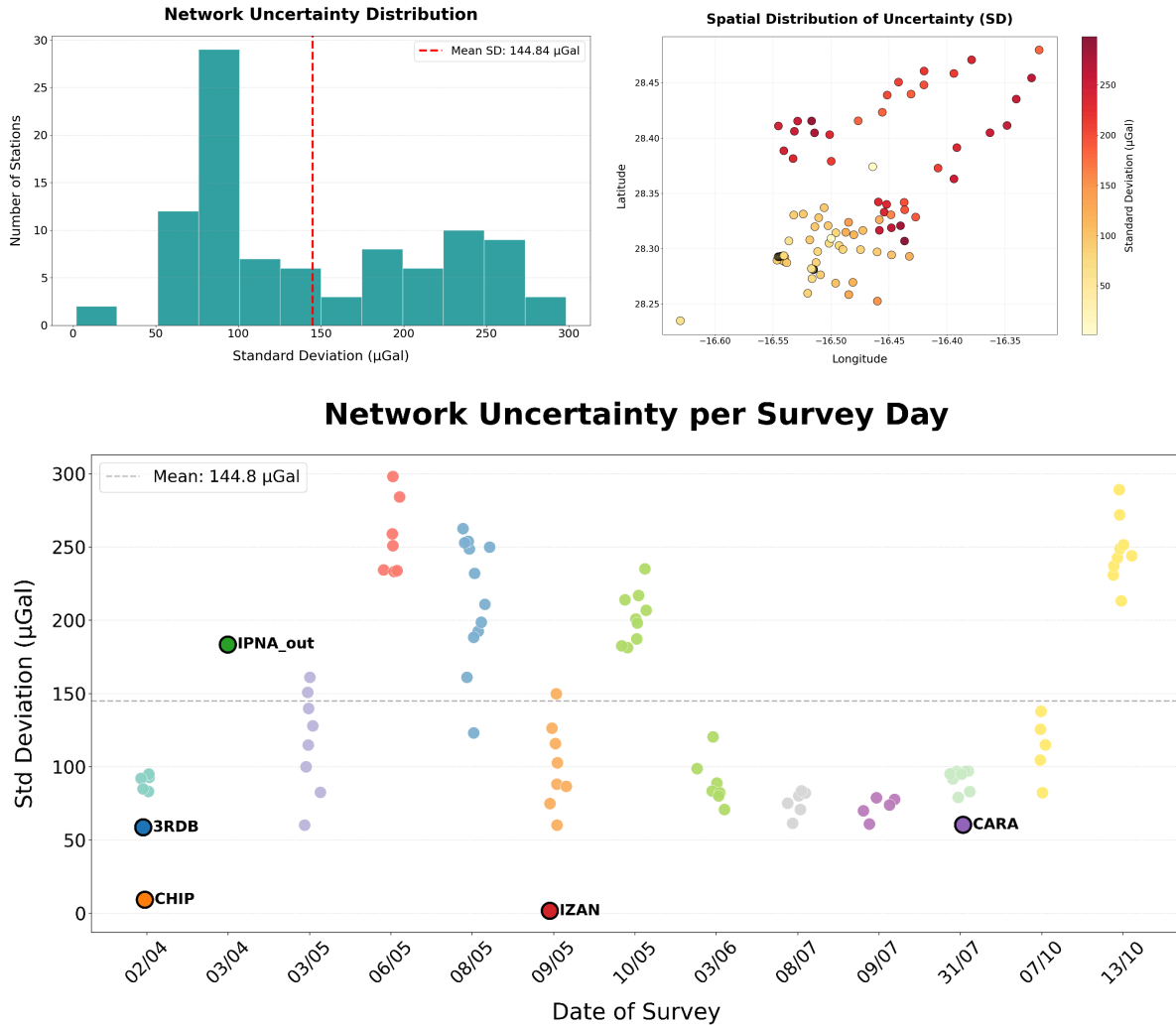
Metadata textfile : */Results_files/GSadjust_MetadataText_20260508-1248.txt*

In the resulting CSV file, the coordinates are rounded at 6 decimal so we loose accuracy (8 decimal) so we re put the original precise coordinate and create a FINAL PRECISE FILE OF THE ABSOLUTE GRAVITY VALUES : *GSadjust/Absolute_Gravity_Final.csv*

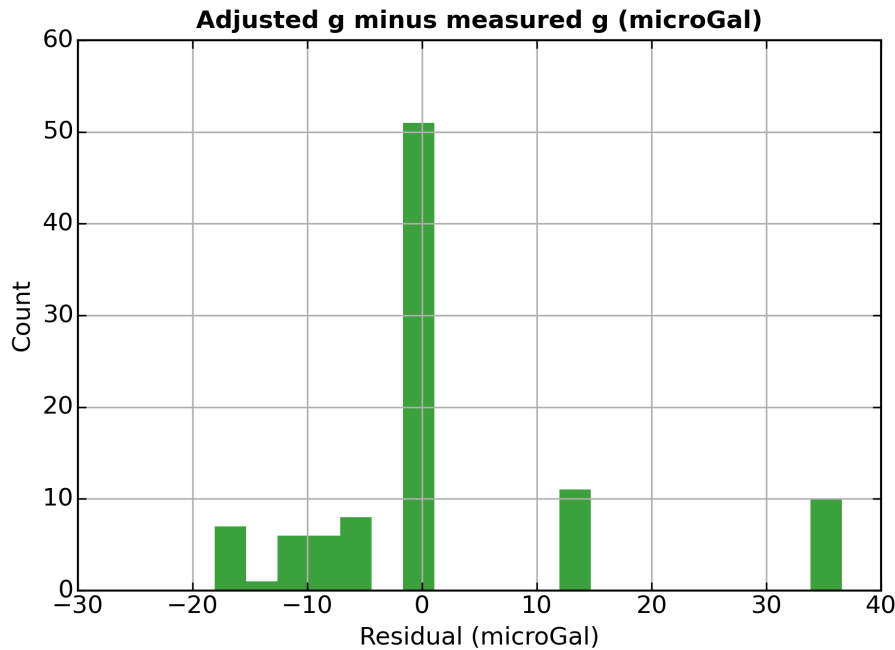
Validation - statistical and physical analysis

The absolute gravity results obtained during this survey were validated through a rigorous statistical and physical analysis, ensuring the data's reliability for subsequent geophysical interpretation. The network adjustment using the least-squares method was

confirmed by the **Chi-square test**, with an 'Accepted' status signifying the strong internal consistency of the mathematical model. The high precision of the survey is demonstrated by a mean **Standard Deviation (SD)** of **144.8 μGal** , with the majority of stations exhibiting uncertainties below 100 μGal (fig histogram uncertainty) and all less than 0.3 mGal.



Analysis : The higher uncertainty are on loops 2, 4 & 10, which are also open loops (lines related to only 1 ref station). The ref stations have low uncertainties.



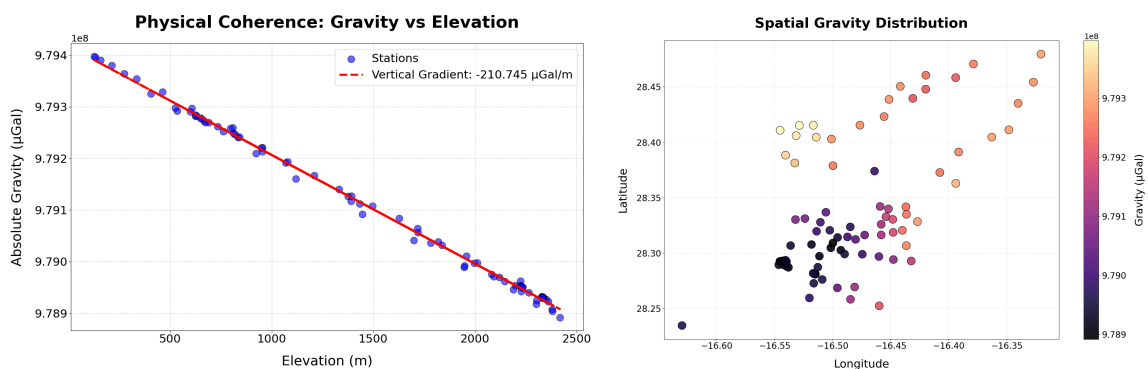
Residual histogram :

Residuals represent the difference between the raw field measurements and the final values calculated by the mathematical adjustment. A histogram of these values illustrates the overall quality of the survey, where a tight distribution centered on zero proves that the network is consistent and free of major systematic errors. The analysis of closure residuals, which are centered at zero with maximum deviations not exceeding **37 μGal** , confirms the effectiveness of the instrumental drift and tare corrections applied.

The residuals at 0 are the open loops that are only linked by one end (line), so they only rely on 1 calculation so nothing to compare to so residual=0. The others are the real closed loops and the residual is still small ($< 37 \mu\text{Gal}$).

From a **physical** point of view, the validity of the measurements is checked by the correlation between absolute gravity and elevation (figure). The network follows a vertical gradient of $-210.7 \mu\text{Gal/m}$, a characteristic value for dense volcanic edifices and highly consistent with the geological context of Tenerife.

These plots and analysis further validate the robustness of the network and the absence of any major systematic biases, also confirming the effectiveness of the instrumental drift and tare corrections applied.



Station	Ref. Height	Ref. Value (Original)	Ref. Value (Ground)	Calculated (Ground)	Difference (μ Gal)
3RDB	72cm	978 962 965.2	978 963 206.3	978 962 797.9	-668.3

The difference between the historical IGN reference and our calculated value is approximately 0.6 mGal. Retrieving a value within less than 1 mGal of a national benchmark after a complex relative survey proves that the network tie is robust and the drift/calibration corrections were successful.

Documentation & links:

Software - [GSadjust | U.S. Geological Survey](#)

USGS, Kennedy et al, 2021 : <https://doi.org/10.3133/tm2D4> - [Techniques and Methods 2–D4: Procedures for Field Data Collection, Processing, Quality Assurance and Quality Control, and Archiving of Relative- and Absolute-Gravity Surveys](#)

User guide v1 - [GSadjust](#)

Manual v2 - [/GSadjust_preprocessing/software_gsadjust_2.0.10/Manual](#)

Base Stations : [Consulta los valores de la gravedad absoluta - Instituto Geográfico Nacional](#)

Problems

- Missing Calibration Factor so need to calculate it from the software but it works well in the end.
- Difficulties to understand what was going wrong : combine Dial & ObsG, apply tares, repartition in the right loops, understand the options we have control on and select the best, know which stations to exclude (3RDB),